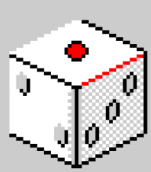


FLEX SENSOR

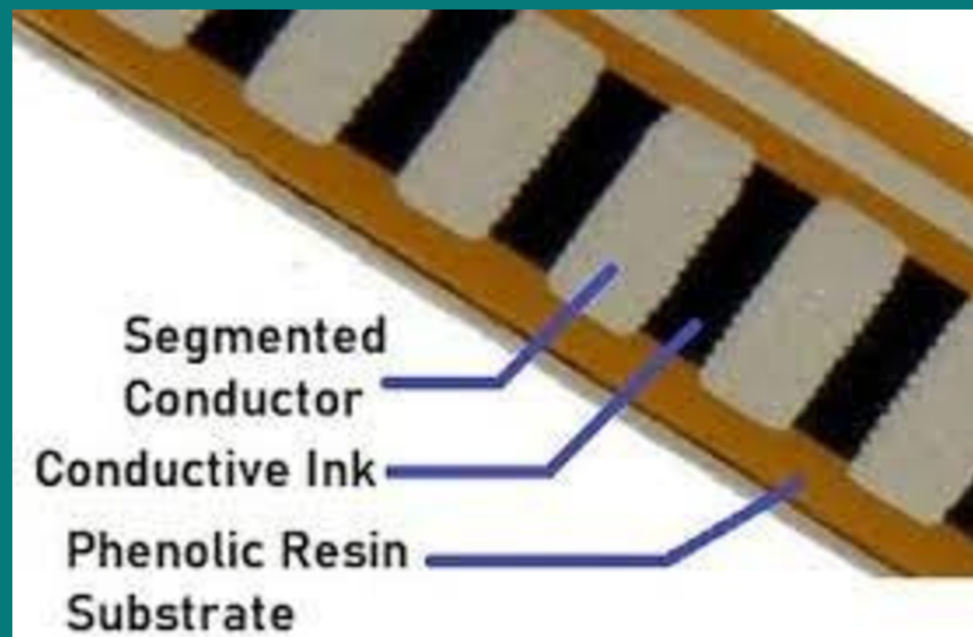
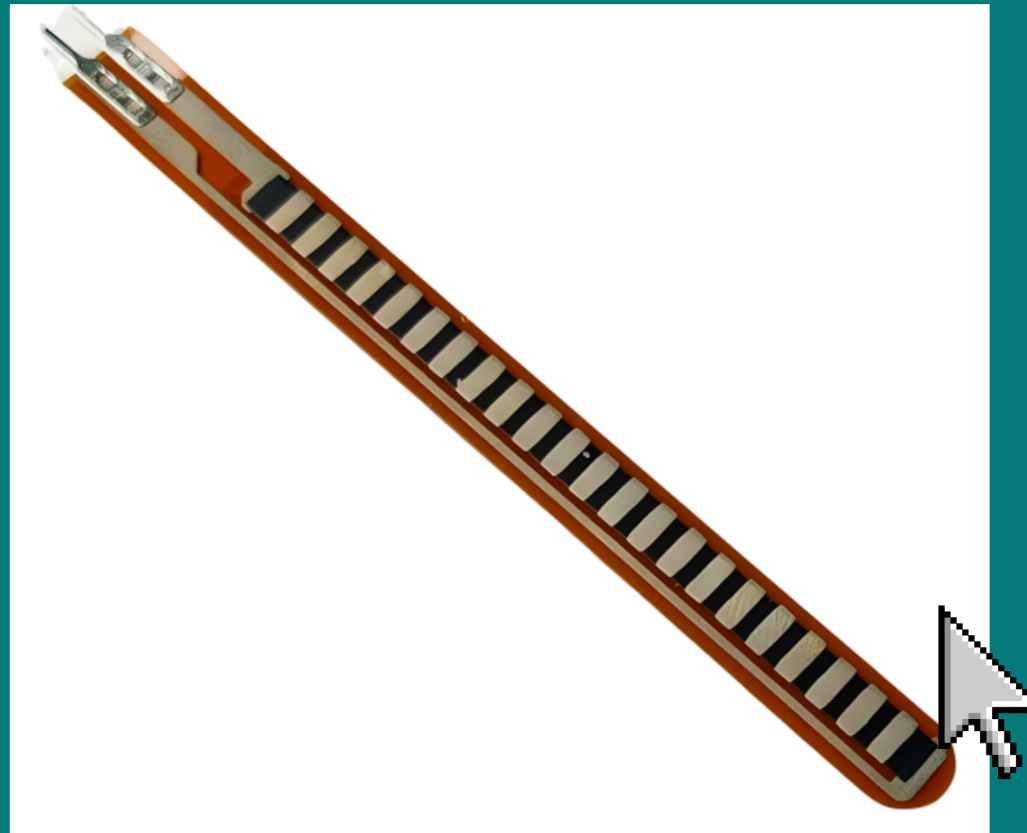


By Meghan K



11:11PM

WHAT is it?



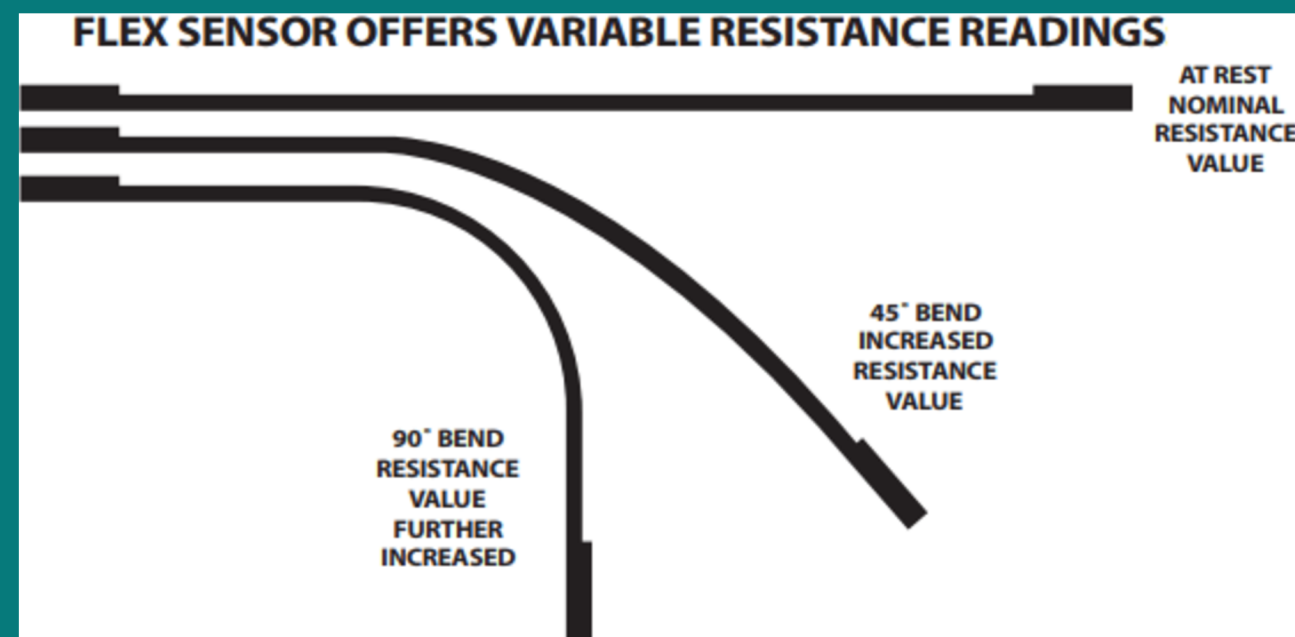
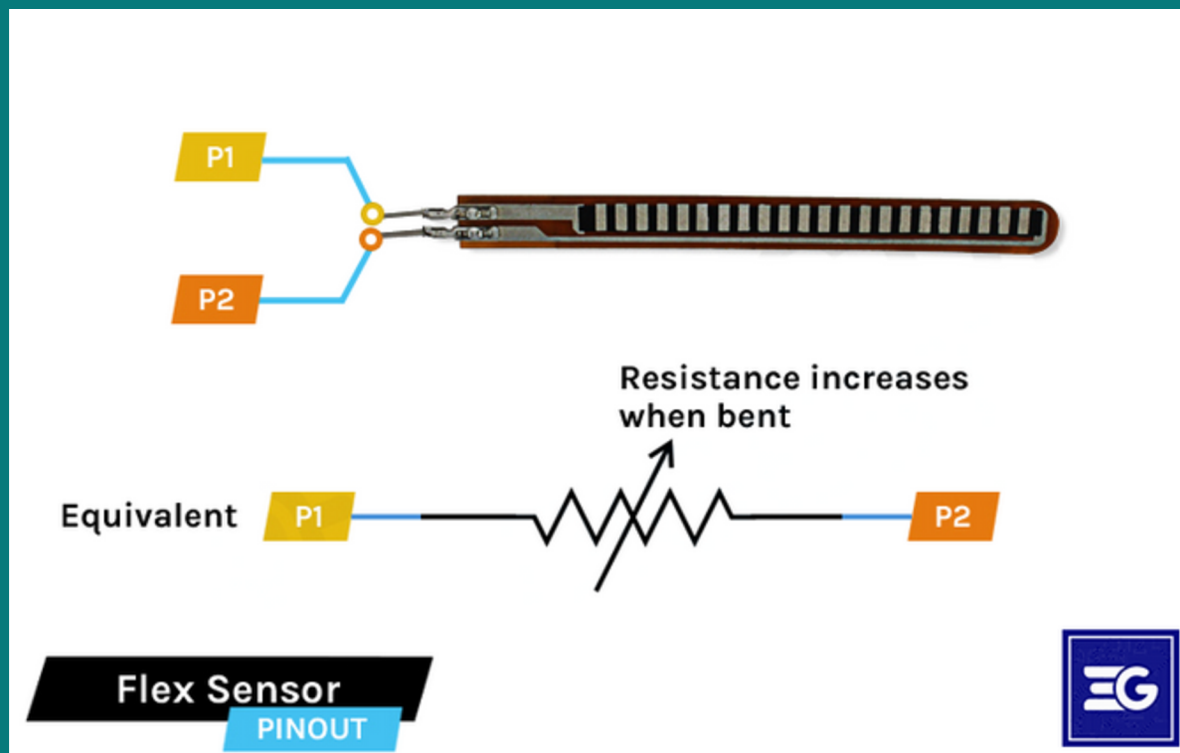
A flex sensor or bend sensor measures the amount of deflection or bending. It is related to force

Substrate: base layer made of bendable material like plastic (polyester)/resin.

Resistive Material: A carbon-based ink or conductive polymer is printed or applied to the substrate. This is essential because the length of the resistive material changes as we bend the sensor. The more distance to traverse the more energy is lost.

Conductive Traces: to conduct electrical signals between the sensor and the external circuit.

WHAT is it?



These sensors can be around 2.5 and 4.5 with little change in functionality. Only difference is the resistance

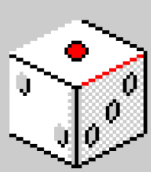
2 Pins :

PIN-1: Connected to positive of power supply.

PIN-2: Connected to ground.

It is not polarised so we don't have to worry about any errors in the direction of the current like with diodes.

I've included a png showing nominal resistance changing as the sensor bends. At 45° the resistance is doubled. At 90° the resistance could be four times (approx.) Variable resistance change the flow of electricity in the given circuit which is why flex sensors are also considered potentiometers. (audio tapers)

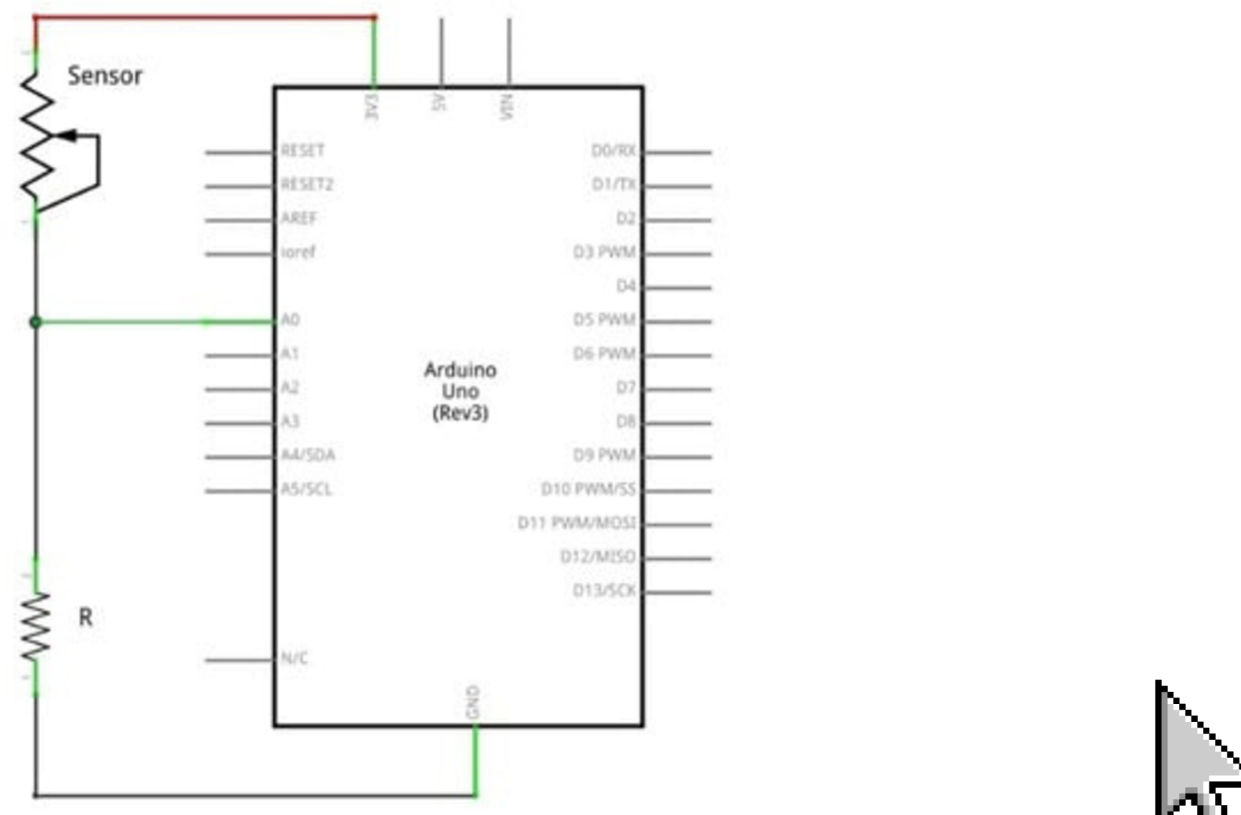
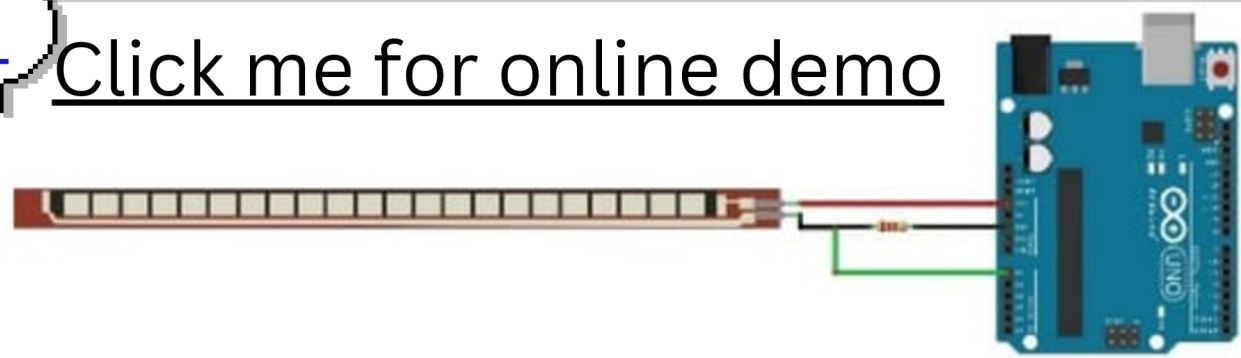


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How it works



[Click me for online demo](#)



ARDUINO/MICROCONTROLLER

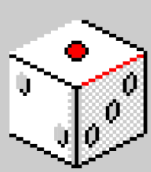
Reads Analog value
Analog()

Serial Communication
Protocol in MAX MSP

control parameters like volume,
cutoff etc.

Scale appropriately

Analog Value:- The change in resistance can be converted into an electrical signal, typically a voltage, using a voltage divider circuit. This voltage can then be read by a microcontroller or an interface device. You usually need a resistor to form a voltage divider, which converts the resistance change to a voltage change.



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Features

FLEX SENSOR Features and Specifications

- Operating voltage of FLEX SENSOR: 0-5V
- Can operate on LOW voltages
- Power rating : 0.5Watt (continuous), 1 Watt (peak)
- Life: 1 million
- Operating temperature: -45°C to +80°C
- Flat Resistance: 25K Ω
- Resistance Tolerance: $\pm 30\%$
- Bend Resistance Range: 45K to 125K Ohms(depending on bend)



TERMINOLOGY

- Impedance: is a measure of how much a circuit resists the flow of alternating current (AC) or varying signals.
- Flat resistance: refers to the resistance measured across a flat surface

Start

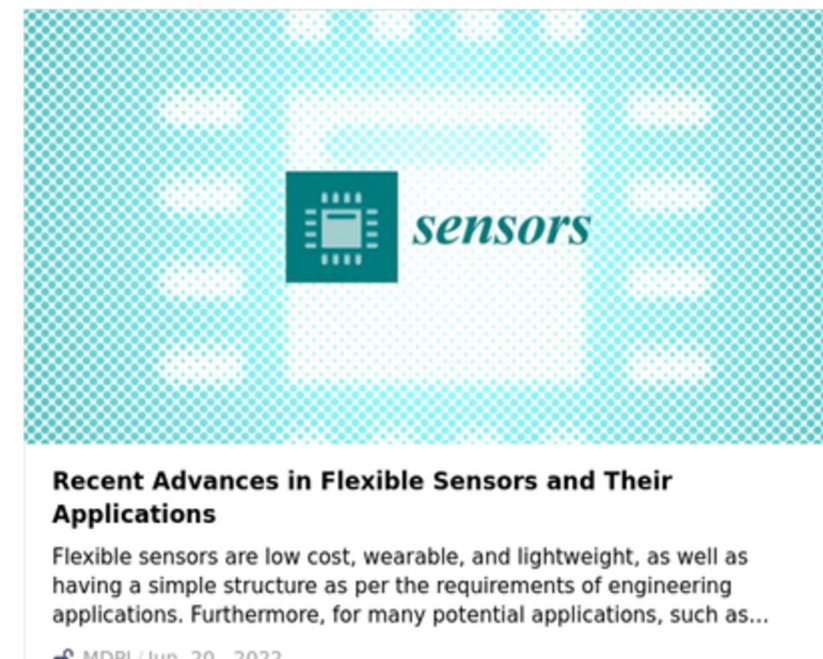
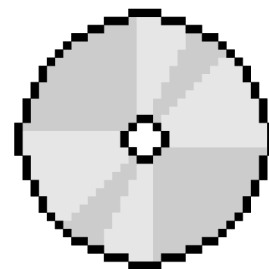
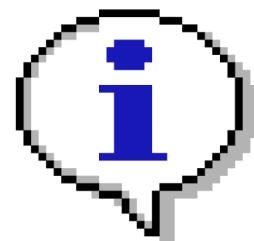
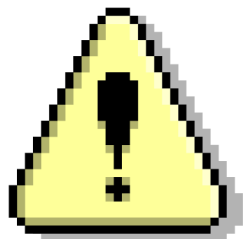
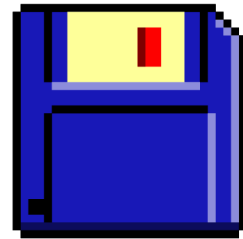


WARNING!

FLEX SENSORS CAN ONLY BE BENT IN ONE
DIRECTION.



Resource Page



<https://www.sparkfun.com/datasheets/Sensors/Flex/FlexSensor.pdf>

[Elprocus.com](https://www.elprocus.com)

